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*
*      STAAD.Pro
*      Version 2007   Build 04
*      Proprietary Program of
*      Research Engineers, Intl.
*      Date=   JAN 18, 2013
*      Time=   17:31:16
*
*      USER ID:
*****
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1. STAAD SPACE
INPUT FILE: 20121120 T-Goal Post 045 FR2.STD
2. START JOB INFORMATION
3. ENGINEER DATE 12-MAY-12
4. END JOB INFORMATION
5. INPUT WIDTH 79
6. UNIT METER KN
7. JOINT COORDINATES
8. 2 5 1 -5.73; 4 5 4.495 -5.73; 5 5 4.495 -4.98; 6 5 4.495 -6.48
9. MEMBER INCIDENCES
10. 3 2 4; 4 4 5; 5 6 4
11. DEFINE MATERIAL START
12. ISOTROPIC STEEL
13. E 2.05E+008
14. POISSON 0.3
15. DENSITY 76.8195
16. ALPHA 1.2E-005
17. DAMP 0.03
18. END DEFINE MATERIAL
19. MEMBER PROPERTY JAPANESE
20. 4 5 TABLE ST H428X407X20
21. 3 TABLE ST H428X407X20
22. CONSTANTS
23. BETA 90 MEMB 3
24. MATERIAL STEEL ALL
25. SUPPORTS
26. 2 FIXED
27. ***** DEAD LOAD *****
28. LOAD 1 LOADTYPE NONE TITLE DL
29. SELFWEIGHT Y -1
30. MEMBER LOAD
31. 5 CON GY -19.6 0.35
32. 4 CON GY -19.6 0.4
33. ***** FRICTIONAL LOAD *****
34. LOAD 2 LOADTYPE NONE TITLE FL
35. MEMBER LOAD
36. 5 CON GX -5.88 0.35
37. 4 CON GX -5.88 0.4
38. ***** WIND LOAD X-DIRECTION*****
39. LOAD 3 LOADTYPE NONE TITLE WLX
40. MEMBER LOAD
41. 4 5 UNI GX -0.364
42. 3 UNI GX -0.364
43. ***** WIND LOAD Z-DIRECTION*****
44. LOAD 4 LOADTYPE NONE TITLE WLZ
45. MEMBER LOAD
46. 3 UNI GZ -0.346
47. ***** NOTIONAL LOAD X-DIRECTION*****
48. LOAD 5 LOADTYPE NONE TITLE NLX
49. SELFWEIGHT X -0.015
50. MEMBER LOAD
51. 5 CON GX -0.294 0.35
52. 4 CON GX -0.294 0.4
53. *****NOTIONAL LOAD Z-DIRECTION*****
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54. LOAD 6 LOADTYPE NONE TITLE NLZ
55. SELFWEIGHT Z -0.015
56. MEMBER LOAD
57. 5 CON GZ -0.294 0.35
58. 4 CON GZ -0.294 0.4
59. ***** LOAD COMBINATION - SERVICE *****
60. LOAD COMB 7 1.0DL+1.0FL
61. 1 1.0 2 1.0
62. LOAD COMB 8 1.0DL+1.0FL+1.0WLX
63. 1 1.0 2 1.0 3 1.0
64. LOAD COMB 9 1.0DL+1.0FL+1.0WLX
65. 1 1.0 2 1.0 4 1.0
66. LOAD COMB 10 1.0DL+1.0FL+1.0NLX
67. 1 1.0 2 1.0 5 1.0
68. LOAD COMB 11 1.0DL+1.0FL+1.0NLZ
69. 1 1.0 2 1.0 6 1.0
70. ***** LOAD COMBINATION - ULTIMATE *****
71. LOAD COMB 12 1.0DL+1.4FL
72. 1 1.0 2 1.4
73. LOAD COMB 13 1.4DL+1.4FL
74. 1 1.4 2 1.4
75. LOAD COMB 14 1.0DL+1.4FL+1.4WLX
76. 1 1.0 2 1.4 3 1.4
77. LOAD COMB 15 1.0DL+1.4FL+1.4WLZ
78. 1 1.0 2 1.4 4 1.4
79. LOAD COMB 16 1.0DL+1.4FL+1.4NLX
80. 1 1.0 2 1.4 5 1.4
81. LOAD COMB 17 1.0DL+1.4FL+1.4NLZ
82. 1 1.0 2 1.4 6 1.4
83. PERFORM ANALYSIS
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PROBLEM STATISTICS

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 4/ 3/ 1

SOLVER USED IS THE IN-CORE ADVANCED SOLVER

TOTAL PRIMARY LOAD CASES = 6, TOTAL DEGREES OF FREEDOM = 18

84. LOAD LIST 7 TO 11
85. PRINT JOINT DISPLACEMENTS ALL

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
2	7	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	8	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	9	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	10	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	11	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
4	7	-0.2099	-0.0023	0.0000	0.0000	0.0000	0.0009
	8	-0.2282	-0.0023	0.0000	0.0000	0.0000	0.0010
	9	-0.2099	-0.0023	-0.0030	0.0000	0.0000	0.0009
	10	-0.2225	-0.0023	0.0000	0.0000	0.0000	0.0009
5	7	-0.2099	-0.0023	-0.0045	0.0000	0.0000	0.0009
	8	-0.2105	-0.0040	0.0000	0.0000	0.0000	0.0009
	9	-0.2288	-0.0040	0.0000	0.0000	0.0000	0.0010
	10	-0.2105	-0.0033	-0.0030	0.0000	0.0000	0.0009
	11	-0.2231	-0.0040	0.0000	0.0000	0.0000	0.0009

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6	7	-0.2105	-0.0040	0.0000	0.0000	0.0000	0.0009
	8	-0.2288	-0.0040	0.0000	0.0000	0.0000	0.0010
	9	-0.2105	-0.0048	-0.0030	0.0000	0.0000	0.0009
	10	-0.2231	-0.0040	0.0000	0.0000	0.0000	0.0009
	11	-0.2105	-0.0053	-0.0045	0.0000	0.0000	0.0009

***** END OF LATEST ANALYSIS RESULT *****

86. PRINT SUPPORT REACTION ALL

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
2	7	11.76	53.04	0.00	0.00	0.00	-41.10
	8	13.58	53.04	0.00	0.00	0.00	-45.23
	9	11.76	53.04	1.21	2.11	0.00	-41.10
	10	12.56	53.04	0.00	0.00	0.00	-43.63
	11	11.76	53.04	0.00	2.53	0.00	-41.10

***** END OF LATEST ANALYSIS RESULT *****

87. LOAD LIST 12 TO 17

88. PRINT MEMBER FORCES ALL

MEMBER END FORCES STRUCTURE TYPE = SPACE

ALL UNITS ARE -- KN METE (LOCAL)

MEMBER	LOAD	JT	AXIAL	SHEAR-Y	SHEAR-Z	TORSION	MOM-Y	MOM-Z
3	12	2	53.04	0.00	16.46	0.00	-57.54	0.00
	4		-43.36	0.00	-16.46	0.00	0.00	0.00
13	2		74.26	0.00	16.46	0.00	-57.54	0.00
	4		-60.70	0.00	-16.46	0.00	0.00	0.00
14	2		53.04	0.00	19.01	0.00	-63.33	0.00
	4		-43.36	0.00	-17.23	0.00	0.00	0.00
15	2		53.04	1.69	16.46	0.00	-57.54	2.96
	4		-43.36	0.00	-16.46	0.00	0.00	0.00
16	2		53.04	0.00	17.58	0.00	-61.08	0.00
	4		-43.36	0.00	-17.37	0.00	0.00	0.00
17	2		53.04	1.11	16.46	0.00	-57.54	3.54
	4		-43.36	-0.91	-16.46	0.00	0.00	0.00
4	12	4	0.00	21.68	-8.23	0.00	3.29	8.62
	5		0.00	0.00	0.00	0.00	0.00	0.00
13	4		0.00	30.35	-8.23	0.00	3.29	12.07
	5		0.00	0.00	0.00	0.00	0.00	0.00
14	4		0.00	21.68	-8.61	0.00	3.44	8.62
	5		0.00	0.00	0.00	0.00	0.00	0.00
15	4		0.00	21.68	-8.23	0.00	3.29	8.62
	5		0.00	0.00	0.00	0.00	0.00	0.00
16	4		0.00	21.68	-8.69	0.00	3.47	8.62
	5		0.00	0.00	0.00	0.00	0.00	0.00
17	4		0.46	21.68	-8.23	0.00	3.29	8.62
	5		0.00	0.00	0.00	0.00	0.00	0.00

	4	0.00	21.68	-8.23	0.00	-3.29	-8.62
13	6	0.00	0.00	0.00	0.00	0.00	0.00
	4	0.00	30.35	-8.23	0.00	-3.29	-12.07
14	6	0.00	0.00	0.00	0.00	0.00	0.00
	4	0.00	21.68	-8.61	0.00	-3.44	-8.62
15	6	0.00	0.00	0.00	0.00	0.00	0.00
	4	0.00	21.68	-8.23	0.00	-3.29	-8.62
16	6	0.00	0.00	0.00	0.00	0.00	0.00
	4	0.00	21.68	-8.69	0.00	-3.47	-8.62
17	6	0.00	0.00	0.00	0.00	0.00	0.00
	4	0.46	21.68	-8.23	0.00	-3.29	-8.62

***** END OF LATEST ANALYSIS RESULT *****

89. LOAD LIST ALL

90. PARAMETER 1

91. CODE BS5950

92. BEAM 2 ALL

93. CHECK CODE ALL

STAAD.Pro CODE CHECKING - (BSI)

PROGRAM CODE REVISION V2.12_5950-1_2000

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
3 ST	H428X407X20	PASS	ANNEX I.1	0.083	14
		53.04 C	63.33	0.00	-
4 ST	H428X407X20	PASS	BS-4.2.3-(Y)	0.022	13
		0.00	3.29	12.07	0.00
5 ST	H428X407X20	PASS	BS-4.2.3-(Y)	0.022	13
		0.00	-3.29	-12.07	0.75

***** END OF TABULATED RESULT OF DESIGN *****

94. FINISH

***** END OF THE STAAD.Pro RUN *****

**** DATE= JAN 18,2013 TIME= 17:31:17 ****

* For questions on STAAD.Pro, please contact *

* Research Engineers Offices at the following locations *

* * * *

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* USA:	+1 (714)974-2500	support@bentley.com	*
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